

Mingzhu Yan

+86 132-7794-9509 | pearlyan919@gmail.com | github.com/pearlyan919

RESEARCH INTERESTS

AI Security and Trustworthy Machine Learning; Large Language Model Security and Safety; Adversarial Machine Learning; Secure Evaluation and Auditing of AI Systems; Systems and Software Security.

RESEARCH EXPERIENCE

Independent Research - Security Evaluation and Robustness Analysis of Large Language Models | Apr 2026 - Present

- Conduct independent research on LLM security evaluation, jailbreak testing, adversarial prompting, cross-model transfer, and output-level safety auditing; release scripts, raw results, figures, and technical reports on GitHub.
- Run controlled GCG experiments on AdvBench behaviors and construct cross-model transfer evaluations across Llama and Gemma model families.
- Perform token-level truncation and output re-auditing with rule-based refusal categories; implement reproducible experiments on an 8 GB GPU with model-specific resource settings.

People's Hospital of Hubei Province - AI Security Researcher / Software Engineer | May 2022 - Present

- Designed a multimodal intent-verification framework for medical AI agents using text, voice, and image inputs with FHIR R4 data to detect potentially unauthorized tool calls.
- Led security-architecture audits of core hospital systems (HIS/PACS) and integrated defensive controls into clinical workflows with sub-second risk response.
- Developed a feature-disentanglement framework for digital pathology to improve cross-institution generalization.

SECURITY ENGINEERING EXPERIENCE

Wuhan Douyu Network Technology Co., Ltd. - Client Security Engineer | Jul 2020 - May 2022

- Worked as an attacker-side analyst in an anti-cheat team, reverse-engineering cheat toolchains and analyzing client security protections.
- Led the native C++ architecture of a core security-authentication library and restructured JNI serialization, reducing latency from 18 ms to 70 microseconds.
- Built Python and Frida tooling for dynamic analysis of obfuscated and anti-debug-protected binaries.

EDUCATION

Wuhan University - M.S. in Computer Technology | Sep 2018 - Jul 2020 | Weighted Average 86.57/100; Second-Class Academic Scholarship; Outstanding Graduate Student.

Northeastern University - B.S. in Computer Science and Technology | Sep 2014 - Jul 2018 | GPA 3.72/5.00; top 11.73%; Second-Class University Scholarship.

PUBLICATION

Yan, M. (First Author). "Decoding the past and future of distant metastasis from papillary thyroid carcinoma: a bibliometric analysis from 2004 to 2023." *Frontiers in Oncology*, SCI, JCR Q2.

TECHNICAL SKILLS

AI Security: PyTorch, HuggingFace Transformers, LLM evaluation, adversarial testing, jailbreak auditing, RLHF/DPO concepts.

Security Engineering: C++, JNI/NDK, IDA Pro, Frida, reverse engineering, obfuscated and anti-debug-protected binaries.

Engineering and Data: Python, Linux, Git, large-scale data cleaning, CiteSpace, VOSviewer, knowledge-graph analysis.

CERTIFICATION

CISSP - Certified Information Systems Security Professional